Introduction to Derivatives

Assignment 1

1) The current level of the S&P 500 is $1040. The risk-free interest rate per year is 2%. Assume negligible dividends. The 6-month futures contract is trading at $1060.
   a) Is there an arbitrage opportunity? Explain.
   b) If there is an arbitrage opportunity, what strategy would you use to exploit it, without using any funds of your own? 2.5 points

2) The price of gold is currently $1,000 per ounce. The forward price for delivery in 1 year is $1,200. An arbitrageur can borrow money at 10% per annum. What should the arbitrageur do? Assume that the cost of storing gold is zero and that gold provides no income. 2.5 points

3) A forward contract on an asset is written at time zero. There are M periods until delivery. Say the carrying charge in period k is q*S(k), where S(k) is the spot price of the asset in period k. Also, say that d(0,M) is the discount factor from end of the last time period to time zero. Consider a portfolio that pays all carrying costs by selling a fraction of the asset as required.

   What is the Forward price? 5 points
